

Assignment -1

1. Security Operations & Digital Forensics

- Q1. What is Network Reconnaissance?
- Q2. What is SIEM?
- Q3. What is Network Traffic Capture?
- Q4. What is Evidence Collection in Digital Forensics
- Q5. Explain Windows Forensics Artifacts
- Q6. Explain Digital Forensics Methodology
- Q7. Explain Network, Cloud & Mobile Forensics

Assignment -2

2. Incident Response & Application Security

- Q1. What are the 6 phases of the Incident Response Lifecycle (according to NIST)?
- Q2. What is Incident Correlation?
- Q3. Why is Input Validation important in Web Applications?
- Q4. Difference between Static and Dynamic Code Analysis
- Q5. What is OS Hardening?
- Q6. What is Ransomware Attack — Incident Response Plan
- Q7. What is Application Security Concepts

Assignment -3

3. Regulations, Standards & Compliance Frameworks

- Q1. 5 Core Functions of NIST Cybersecurity Framework
- Q2. Difference between ISO Standards
- Q3. India's IT Act (2000 & 2008 Amendment)
- Q4. What is PCI DSS?
- Q5. What is HIPAA?
- Q6. What is Comparison of Security Frameworks.
- Q7. Discuss SOX and Cybersecurity.